On the Middle Coefficients of Rational q, t-Catalan Polynomials

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Abstract

Let r, s be coprime positive integers and let $g = (r-1)(s-1)/2$. We prove that the part of the rational q, t-Catalan polynomial $C_{r,s}(q,t)$ of total degree $g - \delta$ has a two-variable Schur expansion

$$C_{r,s;\delta}(q,t) = \sum_{a=0}^{\min\{\delta, \lfloor (g-\delta)/2 \rfloor\}} c_{r,s;\delta,a}\, s_{(g-\delta-a,a)}(q,t), \qquad c_{r,s;\delta,a} \in \mathbb{Z}_{\geq 0}.$$

Consequently, the coefficient of $q^j t^{g-\delta-j}$ is independent of j for

$$\delta \leq j \leq g - 2\delta.$$

The proof uses Ma's scalar rational-master character theorem for the algebraic filtration on a finite-dimensional rational Cherednik module. That character is converted into the bigraded Hilbert series of the maximal-ideal associated graded of the spherical module. Each fixed-deficit piece is an $\mathfrak{sl}_2$-module, and its primitive multiplicities are controlled by a quotient generated in positive deficit. We also deduce a primitive Hodge-vanishing statement for the Catalan variety of Galashin and Lam. An appendix records the exact version of Ma's theorem used here and gives a self-contained normalization argument for the two global grading shifts.

1 Introduction

The cases $r = 1$ or $s = 1$ are immediate, since then the rational Catalan polynomial is 1. We therefore assume that $r, s \geq 2$ are coprime and put

$$g = \frac{(r-1)(s-1)}{2}.$$

In the area–dinv convention of the rational shuffle theorem [12], the rational q, t-Catalan polynomial is

$$C_{r,s}(q,t) = \sum_{D \in \mathrm{Dyck}_{r,s}} q^{\mathrm{area}(D)} t^{\mathrm{dinv}(D)}. \tag{1}$$

For $0 \leq \delta \leq g$, define the fixed-deficit slice

$$C_{r,s;\delta}(q,t) = \sum_{A+B=g-\delta} [q^A t^B] C_{r,s}(q,t)\, q^A t^B. \tag{2}$$

Thus every monomial in this slice has total degree $N = g - \delta$.

Our main result gives more than a constant middle interval: it bounds the support of the full two-variable Schur expansion.

Theorem 1.1 (Schur-support bound)**.** *For every $0 \le \delta \le g$, there are nonnegative integers $c_{r,s;\delta,a}$ such that, with $N = g - \delta$,*

$$C_{r,s;\delta}(q,t) = \sum_{a=0}^{\min\{\delta,\lfloor N/2\rfloor\}} c_{r,s;\delta,a}\, s_{(N-a,a)}(q,t). \tag{3}$$

Equivalently, no constituent $s_{(N-a,a)}(q,t)$ occurs when $a > \delta$.

Since

$$s_{(N-a,a)}(q,t) = \sum_{j=a}^{N-a} q^j t^{N-j}, \tag{4}$$

we obtain the following coefficient statement.

Corollary 1.2 (Flat middle)**.** *For every $0 \le \delta \le g$, the coefficient*

$$[q^j t^{g-\delta-j}] C_{r,s}(q,t)$$

is independent of j throughout

$$\delta \le j \le g - 2\delta. \tag{5}$$

The interval is nonempty exactly when $3\delta \le g$.

The analogous full statement in the classical case $C_n(q,t) = C_{n+1,n}(q,t)$ was formulated by the author as [8, Conjecture 5.38]. Thus Corollary 1.2 proves that conjecture and gives its rational extension. Symmetric string decompositions and related formulas were previously obtained in several special settings. The case with one parameter equal to 3 has an explicit formula due to Kaliszewski and Li [9]; generalized q,t-Catalan polynomials of length at most 4 admit symmetric-chain decompositions by Gorsky, Hawkes, Schilling, and Rainbolt [4]; and a symmetric monomial-string formula for general rational parameters was proposed in [7]. The new point here is the uniform primitive-support inequality $a \le \delta$ for every coprime pair.

The proof is algebraic. Let $L_{r/s}$ be the finite-dimensional irreducible polynomial representation of the type-A rational Cherednik algebra, and let

$$R = \mathbf{e} L_{r/s} = L_{r/s}^{S_s}$$

be its spherical part. Ma's rational-master theorem identifies the bigraded character of the algebraic filtration on R with $C_{r,s}$. The Gorsky–Oblomkov–Rasmussen–Shende filtration then converts this character into the Hilbert series of

$$G = \mathrm{gr}_{\mathfrak{m}} R,$$

where $\mathfrak{m} = R_{>0}$. The fixed-deficit subspace

$$G^{(\delta)} = \bigoplus_{\ell \ge 0} G_{2\ell+\delta,\ell}$$

is an actual finite-dimensional $\mathfrak{sl}_2$-module. Its raising operator is multiplication by $p_2/2$. The multiplicity of a string beginning in position a is therefore a homogeneous dimension of $G/(p_2)$. Finally,

$$\mathbb{C}[z_3, \dots, z_s] \twoheadrightarrow G/(p_2), \qquad \deg z_j = (j,1),$$

and every generator z_j, $j \geq 3$, contributes at least one unit of deficit. A monomial of maximal-ideal order a must consequently have deficit at least a.

The rational-master character framework was formulated by Gorsky and Neguţ [6]. The only input not presently available in a refereed publication is the scalar rational-master character in Ma's updated author manuscript [10]. Ma's filtration comparison is now published [11]. Appendix A identifies the exact manuscript version used here, records the logical dependencies, and supplies a unit normalization that fixes both global grading shifts.

The paper is organized as follows. Section 2 states the Cherednik input and fixes filtration conventions. Section 3 converts the character to the $\mathfrak{m}$-adic associated graded. Sections 4 and 5 construct the fixed-deficit $\mathfrak{sl}_2$-module and prove the support bound. Section 6 gives coefficient formulas and examples. Section 7 records the Hodge-theoretic corollary.

2 The scalar Cherednik character

Set $c = r/s$ and let $H_c(S_s, \mathfrak{h})$ be the rational Cherednik algebra of type A_{s-1}, where

$$\mathfrak{h} = \{(x_1, \dots, x_s) \in \mathbb{C}^s : x_1 + \cdots + x_s = 0\}.$$

Let L_c be the unique finite-dimensional irreducible quotient of the polynomial representation, let

$$\mathbf{e} = \frac{1}{s!} \sum_{w \in S_s} w,$$

and put

$$R = \mathbf{e} L_c = L_c^{S_s}.$$

Let h be the Cartan element in the internal Cherednik $\mathfrak{sl}_2$, and write

$$R = \bigoplus_{k \in \mathbb{Z}} R(k)$$

for the h-weight decomposition. We use the standard normalization in which the constant polynomial has weight $-g$; a homogeneous polynomial of ordinary degree u then has weight

$$k = u - g. \tag{6}$$

Let $F_i^{\mathrm{alg}} L_c$ be the algebraic filtration of Gorsky–Oblomkov–Rasmussen–Shende [5], equivalently Ma's algebraic/Kazhdan filtration, and set

$$F_i^{\mathrm{alg}} R = R \cap F_i^{\mathrm{alg}} L_c.$$

Every filtration step is S_s-stable. Since the symmetrizer is an exact projection in characteristic zero,

$$\mathrm{gr}^{\mathrm{alg}} R \cong \mathbf{e}\, \mathrm{gr}^{\mathrm{alg}} L_c$$

as a bigraded vector space.

The character theorem needed below is the following scalar specialization of Ma's rational-master theorem.

Theorem 2.1 (Ma's scalar rational-master character)**.** *For coprime positive integers r, s,*

$$\sum_{i,k} \dim \mathrm{gr}_i^{\mathrm{alg}} R(k)\, x^k y^i = C_{r,s}(xy, x^{-1}y), \tag{7}$$

where

$$\mathrm{gr}_i^{\mathrm{alg}} R(k) = \frac{F_i^{\mathrm{alg}} R(k)}{F_{i-1}^{\mathrm{alg}} R(k)}.$$

Appendix A explains the exact source and normalization of Theorem 2.1. The remainder of the argument uses that theorem, published results of [5], and elementary commutative algebra.

For reference, the three filtrations used in the proof are summarized below.

Notation	Direction	Meaning
$\mathcal{F}^i L_c$	decreasing	modified powers filtration of [5]
$F_i^{\mathrm{alg}} L_c$	increasing	$(\mathcal{F}^i L_c)^\perp$ for the Dunkl form
$\mathfrak{m}^\ell R$	decreasing	powers of the homogeneous maximal ideal of R

3 The maximal-ideal associated graded

The polynomial representation quotient has the form

$$L_c \cong \mathbb{C}[\mathfrak{h}]/I_c$$

for a homogeneous S_s-stable ideal I_c. Exactness of invariants gives

$$R \cong \mathbb{C}[\mathfrak{h}]^{S_s}/(I_c \cap \mathbb{C}[\mathfrak{h}]^{S_s}).$$

For

$$p_j = \sum_{i=1}^{s} x_i^j,$$

we have $p_1 = 0$ on $\mathfrak{h}$, and

$$\mathbb{C}[\mathfrak{h}]^{S_s} = \mathbb{C}[p_2, p_3, \dots, p_s].$$

Thus

$$R \cong \mathbb{C}[p_2, p_3, \dots, p_s]/J \tag{8}$$

for a homogeneous ideal J.

Let

$$\mathfrak{m} = R_{>0}$$

and form the doubly graded algebra

$$G = \mathrm{gr}_{\mathfrak{m}} R = \bigoplus_{u,\ell} G_{u,\ell}, \qquad G_{u,\ell} = \frac{(\mathfrak{m}^\ell)_u}{(\mathfrak{m}^{\ell+1})_u}.$$

Here u is ordinary polynomial degree and ℓ is $\mathfrak{m}$-adic order.

Let $\mathfrak{a} \subseteq L_c$ be the ideal generated by the positive-degree invariant polynomials. Under $R = L_c^{S_s}$,

$$\mathfrak{a} = \mathfrak{m} L_c.$$

Lemma 3.1. *For every $j \geq 0$,*

$$\mathfrak{a}^j \cap R = \mathfrak{m}^j. \tag{9}$$

Proof. The inclusion $\mathfrak{m}^j \subseteq \mathfrak{a}^j \cap R$ is immediate. Conversely, write an element $v \in \mathfrak{m}^j L_c \cap R$ as

$$v = \sum_\nu r_\nu w_\nu, \qquad r_\nu \in \mathfrak{m}^j, \quad w_\nu \in L_c.$$

Let $\rho : L_c \to L_c^{S_s} = R$ be the Reynolds operator. Since the r_ν are invariant, ρ is R-linear. Hence

$$v = \rho(v) = \sum_\nu r_\nu \rho(w_\nu) \in \mathfrak{m}^j.$$

□

Gorsky–Oblomkov–Rasmussen–Shende define the modified decreasing filtration

$$\mathcal{F}^i L_c = \sum_{j \geq 0} \left(\mathfrak{a}^j \cap \bigoplus_{k < 2j-i} L_c(k) \right). \tag{10}$$

Their Proposition 4.5 shows that every $\mathcal{F}^i L_c$ is preserved by the internal $\mathfrak{sl}_2$.

Lemma 3.2. *At Euler weight k,*

$$\mathcal{F}^i R(k) = \mathfrak{m}^{\lfloor (i+k)/2 \rfloor + 1} \cap R(k), \tag{11}$$

where $\mathfrak{m}^j = R$ for $j \leq 0$.

Proof. Each summand in (10) is S_s-stable, so taking invariants commutes with the sum. At weight k, the summand indexed by j occurs precisely when

$$k < 2j - i, \qquad \text{or equivalently} \qquad j > \frac{i+k}{2}.$$

The powers of $\mathfrak{m}$ form a decreasing chain, so the sum is the smallest occurring power, $\mathfrak{m}^{\lfloor (i+k)/2 \rfloor + 1}$. Apply Lemma 3.1. □

The algebraic filtration is defined using the nondegenerate Dunkl form:

$$F_i^{\mathrm{alg}} L_c = (\mathcal{F}^i L_c)^\perp. \tag{12}$$

The form is homogeneous and S_s-invariant. Its restriction to R is nondegenerate: for $v \in R$ and $w \in L_c$, invariance gives $(v, w) = (v, \rho(w))$. Thus a vector in R orthogonal to R is orthogonal to all of L_c and is zero. The same averaging argument gives

$$F_i^{\mathrm{alg}} R = (\mathcal{F}^i R)^{\perp_R}.$$

Consequently,

$$\mathrm{gr}_i^{\mathrm{alg}} R(k) \cong \left(\frac{\mathcal{F}^{i-1} R(k)}{\mathcal{F}^i R(k)} \right)^* \tag{13}$$

as vector spaces. We use only equality of dimensions; no equivariant identification with the dual is needed.

Proposition 3.3. *If $i+k$ is odd, then $\mathrm{gr}_i^{\mathrm{alg}} R(k) = 0$. If $i+k = 2\ell$, then*

$$\dim \mathrm{gr}_i^{\mathrm{alg}} R(k) = \dim G_{g+k,\ell}. \tag{14}$$

Proof. If $i+k = 2\ell$, Lemma 3.2 gives

$$\mathcal{F}^{i-1}R(k) = \mathfrak{m}^{\ell} \cap R(k), \qquad \mathcal{F}^{i}R(k) = \mathfrak{m}^{\ell+1} \cap R(k).$$

By (6), weight k is ordinary degree $u = g+k$, so the quotient in (13) is $G_{g+k,\ell}$. If $i+k$ is odd, the powers appearing for $i-1$ and i coincide. □

We can now rewrite the rational-master character entirely in terms of G.

Proposition 3.4. *One has*

$$C_{r,s}(q,t) = \sum_{u,\ell} \dim G_{u,\ell}\, q^{\ell} t^{g-u+\ell}. \tag{15}$$

In particular, the deficit of the summand indexed by (u,ℓ) is

$$\delta = u - 2\ell. \tag{16}$$

Thus, if

$$G^{(\delta)} = \bigoplus_{\ell \geq 0} G_{2\ell+\delta,\ell}, \tag{17}$$

then

$$C_{r,s;\delta}(q,t) = \sum_{\ell \geq 0} \dim G_{2\ell+\delta,\ell}\, q^{\ell} t^{g-\delta-\ell}. \tag{18}$$

Proof. A monomial $q^A t^B$ on the right side of (7) becomes

$$(xy)^A (x^{-1}y)^B = x^{A-B} y^{A+B}.$$

Hence $k = A - B$ and $i = A + B$. Proposition 3.3 then gives

$$\ell = \frac{i+k}{2} = A, \qquad u = g + k = g + A - B.$$

Solving for B yields $B = g - u + \ell$, which proves (15). The total q,t-degree is $g - u + 2\ell$, giving (16) and (18). □

The absence of negative deficits is also immediate from (8): every element of $\mathfrak{m}^{\ell}$ has ordinary degree at least 2ℓ, because the invariant algebra has no degree-one generator.

4 Fixed deficit and the internal $\mathfrak{sl}_2$

The internal Cherednik $\mathfrak{sl}_2$ has raising operator

$$E = \frac{1}{2}\left(p_2 - \frac{1}{s}p_1^2\right) = \frac{1}{2}p_2, \tag{19}$$

since $p_1 = 0$ on $\mathfrak{h}$, and $[h, E] = 2E$.

Fix δ and write

$$N = g - \delta.$$

For $G_{2\ell+\delta,\ell}$, the Euler weight is

$$k = (2\ell + \delta) - g = 2\ell - N.$$

The relation $i + k = 2\ell$ then gives $i = N$. Summing over all weights produces an actual weight-graded identification

$$\frac{\mathcal{F}^{N-1}R}{\mathcal{F}^{N}R} \cong \bigoplus_{\ell \geq 0} G_{2\ell+\delta,\ell} = G^{(\delta)}. \tag{20}$$

The quotient on the left is $\mathfrak{sl}_2$-stable by [5, Proposition 4.5]; we transport this action to $G^{(\delta)}$.

The following compatibility is the key bridge between the representation-theoretic and commutative-algebraic descriptions.

Lemma 4.1. *Under* (20)*, the raising operator $E = p_2/2$ is multiplication by the initial form of $p_2/2$ in $G = \operatorname{gr}_{\mathfrak{m}} R$. In bidegrees,*

$$E : G_{u,\ell} \longrightarrow G_{u+2,\ell+1}. \tag{21}$$

Proof. At weight $k = 2\ell - N$, the corresponding summand of the left side of (20) is

$$\frac{\mathfrak{m}^{\ell} \cap R(k)}{\mathfrak{m}^{\ell+1} \cap R(k)}.$$

Multiplication by p_2 raises ordinary degree by 2, Euler weight by 2, and maximal-ideal order by at least 1. Modulo $\mathfrak{m}^{\ell+2}$, the induced map is precisely multiplication by the initial form of p_2. Dividing by 2 gives the action of E. □

Every finite-dimensional $\mathfrak{sl}_2$-module is a direct sum of irreducibles. An irreducible whose lowest Euler weight is $-N + 2a$ has weights

$$-N + 2a, -N + 2a + 2, \ldots, N - 2a.$$

By (18), its contribution to the fixed-deficit character is

$$q^a t^{N-a} + q^{a+1} t^{N-a-1} + \cdots + q^{N-a} t^a = s_{(N-a,a)}(q,t).$$

We have therefore proved the preliminary Schur expansion

$$C_{r,s;\delta}(q,t) = \sum_{a=0}^{\lfloor N/2 \rfloor} c_{r,s;\delta,a}\, s_{(N-a,a)}(q,t), \qquad c_{r,s;\delta,a} \in \mathbb{Z}_{\geq 0}. \tag{22}$$

The remaining issue is the support bound $a \leq \delta$.

For a finite-dimensional $\mathfrak{sl}_2$-module V, the quotient V/EV contains exactly the lowest-weight line of each irreducible summand. Applied to $G^{(\delta)}$, this gives

$$c_{r,s;\delta,a} = \dim \left(G^{(\delta)}/EG^{(\delta)} \right)_{2a+\delta,a}. \tag{23}$$

Since E preserves deficit, this is the corresponding component of the global quotient:

$$c_{r,s;\delta,a} = \dim (G/EG)_{2a+\delta,a}. \tag{24}$$

5 The primitive-support bound

The presentation (8) is a filtered surjection for the homogeneous maximal-ideal filtrations. The image of $(p_2, \dots, p_s)^\ell$ is exactly $\mathfrak{m}^\ell$, so passage to associated gradeds gives a bigraded surjection

$$\mathbb{C}[z_2, z_3, \dots, z_s] \twoheadrightarrow G, \qquad \deg z_j = (j, 1), \tag{25}$$

where z_j is the initial form of p_j. By Lemma 4.1, the image of z_2 is $2E$. Quotienting by it yields

$$\mathbb{C}[z_3, z_4, \dots, z_s] \twoheadrightarrow G/EG. \tag{26}$$

A monomial $z_3^{b_3} \cdots z_s^{b_s}$ has maximal-ideal order

$$a = \sum_{j=3}^{s} b_j$$

and ordinary degree

$$u = \sum_{j=3}^{s} j b_j.$$

Its deficit is therefore

$$\delta = u - 2a = \sum_{j=3}^{s} (j-2) b_j \geq \sum_{j=3}^{s} b_j = a. \tag{27}$$

Thus the source of (26) has no component of bidegree $(2a + \delta, a)$ when $a > \delta$. A quotient cannot create a missing homogeneous component, so

$$(G/EG)_{2a+\delta,a} = 0 \qquad (a > \delta).$$

Together with (24), this proves $c_{r,s;\delta,a} = 0$ for $a > \delta$. Substituting into (22) proves Theorem 1.1.

Proof of Corollary 1.2. Let $N = g - \delta$. Every occurring string in (3) begins at some $a \leq \delta$ and ends at $N - a \geq N - \delta$. Hence every such string contains $q^j t^{N-j}$ exactly once whenever

$$\delta \leq j \leq N - \delta = g - 2\delta.$$

The coefficient there is $\sum_a c_{r,s;\delta,a}$, independent of j. □

6 Consequences and examples

The Schur expansion gives the complete coefficient sequence in each deficit.

Corollary 6.1. *For $0 \leq j \leq N = g - \delta$,*

$$[q^j t^{N-j}] C_{r,s;\delta}(q,t) = \sum_{0 \leq a \leq \min\{\delta, j, N-j\}} c_{r,s;\delta,a}. \tag{28}$$

Consequently, every fixed-deficit coefficient sequence is symmetric, weakly increasing toward the center, constant on the interval $[\delta, N - \delta]$ when that interval is nonempty, and weakly decreasing afterward. Moreover,

$$[q^j t^{N-j}] C_{r,s;\delta} - [q^{j-1} t^{N-j+1}] C_{r,s;\delta} = c_{r,s;\delta,j} \tag{29}$$

for $1 \leq j \leq \min\{\delta, \lfloor N/2 \rfloor\}$.

The free polynomial source in (26) also gives a useful upper bound.

Proposition 6.2. *For all $a, \delta \geq 0$,*

$$c_{r,s;\delta,a} \leq [X^a Y^\delta] \prod_{j=3}^{s} \frac{1}{1 - XY^{j-2}}. \tag{30}$$

In particular, nonvanishing implies

$$a \leq \delta \leq (s-2)a. \tag{31}$$

If $s \geq 3$ and $a = \delta$, the only possible source monomial is z_3^δ, so

$$c_{r,s;\delta,\delta} \leq 1. \tag{32}$$

For $s = 2$, the source in (26) is $\mathbb{C}$, and only $a = \delta = 0$ occurs.

Proof. The coefficient on the right side of (30) counts monomials satisfying

$$\sum_{j=3}^{s} b_j = a, \qquad \sum_{j=3}^{s} (j-2) b_j = \delta.$$

The bidegree component of a quotient cannot exceed the corresponding component of its source. The remaining assertions follow from $1 \leq j - 2 \leq s - 2$. □

When the interval in Corollary 1.2 is nonempty, its common coefficient has the representation-theoretic interpretation

$$\sum_{a} c_{r,s;\delta,a} = \dim\bigl(G^{(\delta)}/EG^{(\delta)}\bigr). \tag{33}$$

Example 6.3 (The pair $(3,4)$)**.** *Here $g = 3$, and*

$$C_{3,4}(q,t) = q^3 + q^2 t + q t^2 + t^3 + qt = s_{(3)}(q,t) + s_{(1,1)}(q,t).$$

The primitive positions are $(\delta, a) = (0,0)$ and $(1,1)$.

Example 6.4 (The pair $(3,5)$)**.** *The explicit rank-word formula of Kaliszewski and Li [9] gives*

$$\begin{aligned} C_{3,5}(q,t) &= q^4 + q^3 t + q^2 t^2 + q t^3 + t^4 + q^2 t + q t^2 \\ &= s_{(4)}(q,t) + s_{(2,1)}(q,t). \end{aligned}$$

Thus the deficit-zero slice is one string of length 5, while the deficit-one slice is one string of length 2. The latter begins at $a = \delta = 1$, illustrating the sharpness of the support bound.

Remark 6.5 (A parameter equal to 3)**.** *When the Cherednik denominator is $s = 3$, the primitive source is $\mathbb{C}[z_3]$. Hence*

$$c_{r,3;\delta,a} = 0 \quad \text{unless } a = \delta, \qquad c_{r,3;\delta,\delta} \leq 1.$$

Thus each nonzero deficit slice is a single Schur string. This is the algebraic shadow of the explicit one-string-per-skip formula in [9].

7 Primitive Hodge vanishing

This section is a corollary of Theorem 1.1, not an input to its proof. Galashin and Lam define the Catalan variety

$$X^{\circ}_{r,r+s} = \Pi^{\circ}_{r,r+s}/T,$$

a smooth affine complex variety of dimension $(r-1)(s-1) = 2g$. Their Theorem 1.1 and equation (1.11) identify its mixed Hodge polynomial with the rational Catalan polynomial [2]:

$$C_{r,s}(q,t) = \sum_{K,P} q^{P-K/2} t^{g-K/2} \dim H^{K,(P,P)}(X^{\circ}_{r,r+s}, \mathbb{C}). \tag{34}$$

The variety satisfies curious Lefschetz. There is a class

$$\gamma \in H^{2,(2,2)}(X^{\circ}_{r,r+s})$$

whose cup product sends (K,P) to $(K+2, P+2)$. For a class in $H^{K,(P,P)}$, set

$$\delta = K - P.$$

Its total q,t-degree in (34) is $g-\delta$, and its q-exponent is

$$a = P - \frac{K}{2} = \frac{P-\delta}{2}.$$

Equivalently,

$$P = 2a + \delta, \qquad K = 2a + 2\delta. \tag{35}$$

Cup product by γ sends a to $a+1$, exactly as the Cherednik raising operator does.

Define the primitive-source quotient

$$\mathrm{Prim}^{P,\delta}_{\gamma}(X^{\circ}_{r,r+s}) = \frac{H^{P+\delta,(P,P)}(X^{\circ}_{r,r+s})}{\gamma\, H^{P+\delta-2,(P-2,P-2)}(X^{\circ}_{r,r+s})}, \tag{36}$$

with a missing source group interpreted as zero. By uniqueness of the Lefschetz–Schur string decomposition,

$$c_{r,s;\delta,a} = \dim \mathrm{Prim}^{2a+\delta,\delta}_{\gamma}(X^{\circ}_{r,r+s}). \tag{37}$$

No direct comparison map between this quotient and G/EG is needed: both sets of primitive dimensions are determined by the same Schur expansion of $C_{r,s}(q,t)$.

Above the middle, curious Lefschetz makes cup product by γ surjective from the preceding Hodge position, so the quotient (36) vanishes for $P > g$. In the source range $P \le g$, the support bound $a \le \delta$ is, by (35), equivalent to $P \le 3\delta$. We conclude:

Corollary 7.1 (Primitive Hodge vanishing)**.** *For every P and δ,*

$$\mathrm{Prim}^{P,\delta}_{\gamma}(X^{\circ}_{r,r+s}) = 0 \qquad \textit{whenever } P > 3\delta. \tag{38}$$

Equivalently, writing $K = P + \delta$, the primitive-source quotient in $H^{K,(K-\delta,K-\delta)}$ vanishes whenever

$$K > 4\delta. \tag{39}$$

A The rational-master input and its normalization

The proof uses the scalar character in Theorem 2.1. Because the relevant rational-master manuscript is currently an author version, we record the precise source and the normalization needed here.

A.1 Version and logical dependencies

The arXiv record is arXiv:2407.00971. The version checked for this paper is the 42-page author PDF linked from Ma's research webpage as "more updated than arxiv version," with PDF generation date 6 October 2025:

`https://xinchma.github.io/pdfs/cuspidal_catalan.pdf`

Its SHA-256 digest is

`1a4bc0ca37d9c7121274248d2792ea6800fd25625af224d2b3e91664d78d9e2d`.

The distinction from the older arXiv version matters because the updated manuscript uses the rational-Catalan normalization required in (7).

In the checked version, the route to the scalar theorem is as follows. The rational shuffle theorem is stated as Theorem 5.4, with its equation (76) identifying the scalar spherical coefficient with $C_{r,s}(q,t)$. Theorem 5.5(b) computes the spherical character for the Hodge filtration. Definition 5.1 converts the Hodge filtration and Euler grading into the Kazhdan grading. Propositions 5.9 and 5.10, summarized in Corollary 1.3, transfer the result to the inductive/algebraic filtration for all positive coprime slopes; the all-slope passage uses the slope symmetry recorded in [3, Corollary 1.1]. The filtration identification is also supported by the published paper [11], especially Lemma 3.9 and Theorem 3.16.

The stronger sheaf-level identification in Conjecture 4.6 of the author manuscript is not used. The character calculation proceeds through the equality of equivariant K-classes in Proposition 4.9, together with localization and the dimension comparison in Theorem 5.5.

A.2 Fixing both global grading shifts

The updated manuscript's discussion of extremal Euler degrees fixes the Euler normalization. Since the character has two gradings, we record an additional normalization by the spherical unit; this also fixes the filtration shift.

Suppose that after the Hodge-to-Kazhdan conversion the scalar character were determined only up to a global factor $x^{\alpha}y^{\beta}$. The spherical module R has a unique lowest Euler-weight line, spanned by the constant polynomial. Its Euler weight is $-g$ by the standard normalization [1, Theorem 1.10]; see also [5, Section 2.2.3]. Its algebraic filtration degree is g: in the notation of Proposition 3.3, the unit is $G_{0,0}$, so $k=-g$, $\ell=0$, and $i+k=2\ell$ gives $i=g$. Therefore the unit contributes

$$x^{-g}y^{g}.$$

On the Catalan side, there is a unique area-zero path, and its monomial is t^{g}. Under the substitution $(q,t)=(xy,x^{-1}y)$, this monomial is again

$$(x^{-1}y)^{g}=x^{-g}y^{g}.$$

Thus the unique lowest-weight term agrees on both sides, forcing

$$\alpha = \beta = 0.$$

This completes the normalization of Theorem 2.1 in the form used in the present paper.

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